

Predictive Sustainability Control: A review assessing the potential to transfer big data driven ‘predictive policing’ to corporate sustainability management



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ABSTRACT

This article discusses the potential to transfer big data algorithms developed for ‘predictive policing’ to the field of corporate sustainability. To do so the paper starts with the thought experiment asking if major corporate scandals with disastrous environmental (BP Oil spill) or social (Rana Plaza building collapse) consequences, or global warming and floods could be prevented if big data driven predictive algorithms were in place. The article reviews first efforts to utilize big data for promoting sustainability and for reducing harm. By analogy the concept of ‘predictive policing’ is identified to be transferred to a concept called “Predictive Sustainability Control”. A systematic literature review on predictive policing is conducted. In a next step all parameters, characteristics, functional areas and processes, as well as legal and ethical issues of predictive policing were clustered and presented. Subsequently the concept is developed out of the clustered themes and criteria (Table 1) and a definition is provided: **Predictive Sustainability Control** is the use of analytical techniques to identify subjects for mutual deliberation, supervision and intervention with the goal of preventing future harm related to environmental, social and governance issues, solving past scandals, and identifying potential actors/corporations of unsustainable activities and their stakeholders in the near future. Furthermore three functional areas (sustainability management, stakeholder partnership and regulatory integration) are defined and the concept is operationalized in a big data driven environment (Fig. 1). In the operationalization the concept of a digital ‘planetary nervous system’ is proposed and eXtensible Business Reporting Language data repositories used in corporate data management and Corporate Social Responsibility reporting were integrated to arrive at a data set, where predictive analytics could be applied to prevent future harm and reduce current unsustainability. In conclusion the question of governance, data protection and privacy is discussed critically and future research avenues for theory advancement and empirical testing are presented. In closing limitations of legality and functionality are discussed. The overall scientific value can be seen in the potential, which big data and algorithms have also for promoting and enforcing sustainable development based on rigorous data management leading to the predictive identification of likely unsustainable events.

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1. Introduction

Imagine a world where the 2013 collapse at Rana Plaza building for textile production in Bangladesh killing 1127 workers did not happen. Imagine further that also the Deep Horizon oil spill with 4.9 million barrels of oil leaking into sea did not take place and toxic smog, destructive floods and human rights violations were also under control. Imagine that the prevention of both social

and environmental catastrophes is the result of Big Data driven algorithms used to profile companies and industries according to their risk to run into an unsustainable and socially irresponsible behavior and incidents.

Also imagine that 17 sustainable development goals along with the more specific 169 targets for 2030 adopted by the United Nations 2015 in Paris will not lead to the frustration caused by the aspirations evoked in Kyoto 1997, Copenhagen 2009, Durban 2011 or Warsaw 2013 just to mention a few. Imagine instead measureable and controllable goals and targets united in a

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unified and transparent data repository open not only to regulators, but also to other stakeholders such as NGOs, media and last but not least shareholders and financial analysts.

On a policy level the vision that something changes *in real* was supported by U.S. president Barak Obama in his speech (Roberts, 2015) in Alaska on September 1st 2015 where he clearly positioned himself against climate change deniers and called for coordinated action: “*And the fact is that climate is changing faster than our efforts to address it. [break] That, ladies and gentleman must change. [break] We are not acting fast enough. [...] Even America and China together cannot do this alone. [...] We have to do it together*” What, if this necessary and expectant vision based on technological advancements could become real as predictive algorithms, unified data repositories for the ‘internet of things’ and statistical modeling allows for transparency and coordinated action unseen in the history so far? What if sustainability and corporate social responsibility would be re-defined from a weak and more aspirational exercise into a strong contribution to create a sustainable world, that – also by facts and figures – is sustainable? This transformative aspect of sustainability in business and society, that so many scholars have called for (e.g. Mader, 2013; McCormick et al., 2013 or Higgins, 2013), could be reached applying big data and predictive analytics. In this regard this paper further elaborates primary steps that have been developed recently for bringing together Big Data and sustainability: Zhao et al. (2016) proposed an optimization model for green supply chains, Song et al. (2016) undertook the work of bringing together theories and methods for Big Data driven environmental performance assessment and Du et al. (2016) focused on production optimization through the cap-and-trade system. A more holistic approach can be found in Seele (2016b) discussing a digital surveillance architecture framed within panopticon theory. Other papers outlined the potential and technicalities of big data for cleaner production such as Huang et al. (2016), Li et al. (2016), Zhang et al. (2016) or Louhghalam et al. (2016). Building on these recent pioneer papers in the intersection of Big Data and sustainability, this paper presents – based on a systematic literature review on predictive policing – a conceptual framework given technological advancements and here basically the power of predictive algorithms already used in other contexts such as consumer research, security issues and law enforcement. Big Data and predictive algorithms are used today to predict consumer behavior and purchasing decisions leading to recommendations like ‘customer’s that have bought this book also bought ...’.

One case documented illustrates the power of predictive algorithms in consumer research quite well. Target Corp. operating discount stores in 2012 became famous for an episode as the company sent out customized advertisements for maternity clothing and nursing furniture to a high school girl. It turned out that the company due to profiling purchasing habit formations could know about the pregnancy before the pregnant mother and her family (Hill, 2012). This predictive power building on profiling of individuals was also introduced in the literature (Duhigg, 2012) indicating the potential for businesses by tracing habits and changes in habits of consumers. From a systematic point of view, profiling consumer research is relevant on the micro level of the consumer. Predictive policing instead involves both a micro and meso level, as for the profiling not only data of the individual person from past and present is used, but also context data regarding neighborhood, social network, or previous communications with others. In this regard it is noteworthy that predictive policing was derived from predictive consumer research as applied by Amazon or Wal-Mart. The general outline here is that predictive policing “moves law enforcement from focusing on what happened

to focusing on what will happen and how to effectively deploy resources in front of crime, thereby changing outcomes” (Beck and McCue, 2009: 18). This general notion of predictive policing in turn makes it particularly suitable to be transferred to sustainability, as the idea – as outlines in the two visions above – is to change outcomes. The potential of Big Data for sustainability has already been discussed and promoted. Among the most prominent are Gijzen (2013) in Nature and Helbing on future ICT (2012). Hence, the approach of predictive policing compared to predictive consumer research is considered to be more suitable to be transferred to sustainability. Due to data tractability new levels of transparency and accountability occur also for corporations. This fundamental “time-ontological shift due to 24/7/365 digital transparency” (Seele, 2016b) offers opportunities for the promotion of sustainability both on a corporate level as well as on a regulatory level. Overall sustainability can be increased by *changing outcomes* (here reducing and overcoming unsustainable behavior), which necessarily involves context data such as environmental data or data from external stakeholders and society at large that are affected by unsustainable behavior.

The paper is organized in the following way. First, the state of the research on predictive policing is reviewed systematically and clustered into six guiding themes. For each thematic cluster the core findings are collected and in a second step transferred from *crime prevention* to the concept of Predictive Sustainability Control (PSC). Subsequently the transfer of the core concepts and guiding principles leads to the conceptual framework of PSC, linked to the field of cleaner production and sustainability reporting (here eXtensible Business Reporting Language (XBRL) repositories) as reference fields. The transfer from predictive policing to PSC is organized along the three lines: a. conceptualization based on the review on predictive policing, b. operationalization in the context of corporate sustainability based on XBRL and the idea of an inter-linked ‘planetary nervous system’ and c. advancing theory towards a new paradigm of big data driven sustainability theory. Limitations and the value of the concept are presented in closing. What clearly limits the proposed concept is that predictive policing is mandated and legally governed more clearly. The state has the mandate and monopoly to law enforcement. Sustainability however is up to political debate and still far from common sense or mandatory guidelines. Hence the proposed concept of PSC next to technological challenges has its value more in stimulating debate and adding to the discussion of big data supported sustainability promotion, than offering an architecture of designing an algorithm or generic program to actually facilitate PSC.

2. Review and thematic clustering of the literature on predictive policing

In order to transfer the concept of predictive policing to Predictive Sustainability Control (PSC) for corporate unsustainability a systematic literature review on predictive policing is conducted. Between June and August 2015 the keywords ‘predictive policing’, ‘predictive analysis’ and ‘predictive algorithm’ were searched in the major literature databases including ABI Inform database, Emerald Management Xtra, Business Source Premiere and EconLit databases. Further entries from Google Scholar were included (excluding doubles) to arrive at a list of 73 publications.¹ The sample list then was reviewed and clustered into six themes along

¹ As journal space is limited I suggest to add a footnote here in the case of publication, stating a. that all papers used in the review were marked with an * and b. that the full list of all 73 references can be obtained on request from the corresponding author.

the emergent topics. The six themes then were grouped in a specific order to organize the review from describing and understanding predictive policing via existing cases to legal and ethical issues. This is required as the topic stems from a disruptive technology that challenges normative assumptions and concepts such as privacy and 'presumption of innocence'. The six structured themes read as: 'definitions', 'general scope and models', 'applied scope and existing cases', 'evaluation of techniques', 'legal and ethical considerations', and finally 'pitfalls of predictive policing'.

In the following the literature from each of the six themes is reviewed. Subsequently the theme is discussed regarding the potential to develop a framework (see table at the end of the review) of Predictive Sustainability Control analogous to predictive policing.

2.1. Definitions and overview of predictive policing

In the following, a common understanding is developed from the literature of what predictive policing and predictive algorithms are. Hence, I start with the underlying technology before turning to the specific application context of predictive policing. As predictive algorithms are around and well developed in several contexts such as homeland security (McCue, 2006), law enforcement (Wang et al., 2012), business intelligence such as data mining (McCue, 2006) or consumer behavior (Duhigg, 2012; Beck and McCue, 2009) it is considered helpful in understanding what the underlying predictive capacities are about. As mankind was always fascinated by predicting the future, it is only from the 20th century onwards that prediction has become more a science than "magic" (Malek, 2008). This scientific background being more "science-based and goal-oriented" is based on "machine learning, pattern recognition, data mining and computer technology" (Malek, 2008). One should add here one of the more recent discoveries and debates on machine learning known as the debate around "the internet of things" (Greengard, 2015), as we will see in the conceptual framework.

The overall effect of predictive capacities is described as being more efficient (Kennedy et al., 2011; Greengard, 2012; Camacho-Collados and Liberatore, 2015), the predictive capacities have been implemented in various contexts and have a significant impact on applications ranging from "business, communication systems and politics to health monitoring and environmental protection" (Malek, 2008). Here we can see that predictive algorithms already pave the way for applications in the area of cleaner production and sustainability as environmental topics are already monitored by algorithmic devices.

Out of the several application contexts of predictive algorithms, predictive policing has been chosen as the most suitable one as it has several analog parameters making it a suitable choice to develop Predictive Sustainability Control from here. The most important one is prevention (instead of creation of additional demand as in marketing) of a harmful outcome. Beck and McCue (in a paper on what policing can learn from Amazon (2009)) point to the fact that predictive policing increases the effectiveness by having "the ability to make every agency more efficient, regardless of the availability of resources". This is important as the approach of predictive algorithms is not only science-based but in addition, due to its technological replicability, also effective given restricted resources and the ability of "changing outcomes" (Beck and McCue, 2009: 18). Given the transformative approach of the sustainability concept (Mader, 2013; McCormick et al., 2013 or Higgins, 2013) changing outcomes is of major importance in developing the Predictive Sustainability Control concept. The paradigmatic shift in police work hence is from "focusing on what happened to focusing on what will happen" associated with a statistical likelihood of an event to happen.

The specific background from which predictive policing was developed also shows its suitability to be transferred to sustainability issues. PredPol, a start-up that has developed an algorithm for predictive policing, was developed by two researchers from University of California, Los Angeles, who discovered that "criminal activity and seismic activity follow surprisingly similar patterns" (Huet, 2015: 46). In order to build a preventive system, PredPol adopted the earthquake prediction software by following two types of variables: "a fixed factor (like an earthquake fault or a rowdy bar) or a variable factor (like another earthquake, which causes after-shocks nearby, or a gang shooting, which triggers retaliatory shootings in the same neighborhood). Each factor can be boiled down to the usual rate it triggers other crimes" (Huet, 2015: 46).

One important outcome for developing Predictive Sustainability Control here is the general functioning of adopting algorithms across different fields of application. In the same vein it can be suggested that the Predictive Sustainability Control concept also makes use of fixed and variable factors in operationalizing the sustainability-enforcement mechanism.

Once predictive policing algorithms have been introduced, their applications were manifold in supporting police work. Residential burglary was among the first contexts where predictive policing was used, as geographical data and tagging allows for variables that can be easily accessed from the archives of the police. This approach called 'predictive mapping' has been "extended to other types of crimes such as pedal cycle theft, vehicle crime and street robbery" (Chainey, 2012). Based on these map based application contexts that still were close to the model of earthquake prediction that also made use of geographical data, the idea of "crime forecasting" has further been developed to facilitate "policy and planning decisions" also supporting "tactical deployment of police resources" (Gorr and Harries, 2003: 551). Further development of predictive policing based on geo-data later on was extended by "long- and short-term horizons, univariate and multivariate methods, and fixed boundary versus ad hoc spatial cluster areal units for the space and time series data" (Gorr and Harries, 2003: 551).

Given the accelerated progress in ICT in the most recent past, algorithms became more and more influential and major cities all over the globe make use of predictive algorithms supporting police work. Greengard in a remarkable paper on policing in the future – also based on the principal model developed in Los Angeles – presents a definition, that comprises all relevant aspects in this new approach and that will help at a later point to conceptualize also Predictive Sustainability Control. Greengard defines that predictive policing "utilizes mathematical algorithms and computer programs to assist law enforcement agencies in predicting where crimes will occur" (Greengard, 2012: 19). Next to the general definition, Greengard also describes the basic work flow of pattern recognition used in predictive policing algorithms: Predictive policing "utilizes data concerning the time, distribution, and geography of past criminal activity into a database to determine crime patterns, which allows police to adjust patrols and resources to improve crime deterrence and better apprehend subjects" (Greengard 2012: 19–20).

The increasing prevalence and success of predictive policing following McCue and Parker (2003) goes back to enhance decision-making based on empirical data as a main feature. The practical application however was enabled particularly due to their low implementation thresholds. Among those were low "IT budgets, specialized personnel, or advanced training in statistics" (McCue and Parker, 2003). This means, once technicalities were set up, operating a data mining predictive algorithm does not come at high transaction costs. The algorithm, as demonstrated above by the transfer from earthquake prediction to predictive policing, has one key feature, that makes it a great support, a novelty and a disruptive

technology: the primary use is bottom-up and data-driven to “find something new in the data – to discover a new piece of information that no one knew previously” (McCue and Parker, 2003). As in scientific research the algorithm starts with a hypothesis and then – mining the data – “seeks to verify or refute the hypothesis based on the data”. In return, this means that those working with predictive algorithms do not need ICT and software engineering skills but what is needed instead is what McCue and Parker (2003) call “domain knowledge [...] which means one is in the position to evaluate the value or validity of the results”. For our case of developing Predictive Sustainability Control this means that those operating with a Predictive Sustainability Control algorithm need to be experts in sustainability more than being trained as software engineers.

One of the consequences that predictive policing brings for police officers is that it also transforms everyday police work. Neyfakh points out that due to the predictions based on empirical data, “interrogations could also become less coercive as agencies across the country decide to abandon their traditional interrogation method” (Neyfakh, 2015).

Overall we may state here that predictive policing has a huge impact on the way how crime prevention can be organized and thus we may follow authors like McCue and Parker when stating that predictive policing is a “significant paradigm shift for the police executive” (McCue and Parker, 2003).

2.2. General scope: spatial and/or temporal dimension of the technology

The most important feature of predictive policing in comparison to traditional policing is that so far existing methods of policing are “intrinsically retrospective” (Bowers et al., 2004) and thus adds to the “time-ontological shift” of instant-transparency (Seele, 2016b). Predictive policing on the contrary with the help of big data and predictive algorithms, allows probabilistic pattern recognition to predict future risks regarding policing with geographical data. Predictive policing however depends on decisions to be made by police officers. Hence, predictive policing is understood as “decision support system”, which in a case study outperformed “the patrolling area definitions currently in use” (Camacho-Collados and Liberatore, 2015).

2.2.1. Spatial dimension

As mentioned above the basic idea is that of a map along with data-driven map-evaluation. One consequence also is the focus on “places, not people” (Curry, 2015), as profiling also raises legal and ethical issues as we will see below. Spatial distributions however offer predictive models to make person-independent forecasts, that make police work more effective, also on the level of backup covering (Curtin et al., 2010): Here researchers have found that burglary for example from a data perspective is “communicable with properties within 400 m of a burgled household being at a significantly elevated risk of victimization for up to two months after an initial event” (Bowers et al., 2004: 641). The same applies for shootings, where it could be shown that predictive “risk terrain modeling” compared to retrospective mapping approaches over six months produced significantly more reliable predictions (Caplan et al., 2011). To develop the spatial dimension geographic attributes are needed as well as qualities of space “that connect to crime outcomes and would not be the result of identifying individuals from specific groups or characteristics of people as likely candidates for crime, a tactic that has led police agencies to be accused of profiling” (Kennedy et al., 2011: 339).

2.2.2. Temporal dimension

Whereas retrospective mapping is based on empirical data from the past, predictive policing forecasts probable future events based on predictive algorithms fed with spatial data. The important temporal dimension in predictive policing however is: how far into the future? Here a study from Cohen et al. (2007) developed a micro level scale of a one month ahead forecast. In their model an area is divided into 141 square grid cells consisting of 4000 feet each on a side. The period used is eight years from the past with a forecast horizon of one month. Technically an indicator model was developed making use of a linear regression model, a neural network and a “proven univariate, extrapolative forecast method” (Cohen et al., 2007: 105). Also in their study, the results are significantly more effective compared to traditional police work.

2.2.3. Spatio-temporal models

Next to the spatial and the temporal dimensions in recent times increasingly spatiotemporal models for decision support are developed. Gerber (2014) introduced a model complemented by a Twitter-driven predictive algorithm. Twitter messages are often “tagged with precise spatial and temporal coordinates” and thus allow to automatically identify discussion topics across a major city” by linguistic analysis and “statistic topic modeling (Gerber, 2014: 115). Another study, also making use of Twitter to arrive at a spatio-temporal model has been presented by Wang et al. (2012). They developed a spatio-temporal modeling making use of geographic, demographic and Twitter-derived information. The model they develop is called the “spatio-temporal generalized additive model (STGAM)” (Wang et al., 2012) and what makes it a very useful resource for developing Predictive Sustainability Control is that the STGAM model also “can be generalized to other applications of event modeling where unstructured text is available” as in the case of the proposed concept here.

On a concluding level the general technology has some features going beyond the temporal-spatial dimensions, that nevertheless are important to understand the concept and to develop it further. Skogan et al. (2003) present a concept that impacts three functional levels such as “police management, criminal justice integration, and community/business partnership” (Skogan et al., 2003). This is particularly of interest as the third level involves external stakeholders, which could also be a major feature for PSC. More precisely their concept engages in the assessment of community needs and facilitating of “easy and convenient information sharing and intelligence gathering from the community” (Skogan et al., 2003).

On the level of technical design Kuo et al. (2013) present a procedural design that could also be employed for developing PSC: They suggest to “(1) geocoding the data, (2) defining the hotspots, (3) organizing the best patrol routes, and (4) estimating the effectiveness” (Kuo et al., 2013: 138). Here it is important to conceptualize the lack of an executive authority for PSC, as the patrolling at the site on the one hand is important, but on the other hand is not feasible for a global PSC. Therefore, it is suggested to adopt the 4-step program to a more data driven solution making use of sustainability data directly from companies, possibly derived from XBRL unified reporting (Seele, 2016b) to obtain relevant information and collaborate with NGOs in order to prevent likely events of unsustainability.

2.3. Applied scopes: specific case studies on predictive policing

In this chapter – as the general concepts and scopes have been introduced – existing application examples are presented. As predictive policing is used on a global scale there are several case studies in the literature, that help translating the concepts into specific empirical data for the successful use of predictive policing.

Amongst those case studies, those cases are looked after, that help in developing the concept of PSC.

The review of the literature resulted in a collection of scholarly articles presenting case studies from different parts of the world where predictive policing is applied. Amongst them the majority of cases stems from the U.S.A. from cities such as Los Angeles (Ashley, 2006; Stomakhin et al., 2011), Chicago (Dugato, 2013), New York (Ashley, 2006), Minneapolis (Egge, 2011), Philadelphia (Haberman and Ratcliffe, 2012) or Santa Cruz (Huet, 2015). From Europe case studies from Madrid (Camacho-Collados et al., 2015) and Trafford, Greater Manchester (Chainey, 2012) were among the reviewed papers and one article deals with the introduction of a predictive policing system in Trinidad and Tobago (Norton, 2013). In this regard it is worth noting that the different countries approach the topic of predictive policing from different angles. Whereas in the U.S. the topic is discussed debating and advancing the state of the art and the interlinkages of start-up companies with police authorities (Huet, 2015), the article from Trinidad and Tobago takes a different route. Norton makes the case for the necessity in the light of “current volume of crime being committed” to advance the case for “implementing data mining” technologies. Norton makes it very clear, that the motivating force is hope, “that this technology will provide decision makers with intelligence from the crime data to inform their strategic planning” (Norton, 2013: 32).

The reviewed case studies analyze specific programs and their application. Amongst the predictive programs presented are CLEAR (Ashley, 2006; Skogan et al., 2003), I-Clear (Dugato, 2013), COMP-STAT (Bratton and Malinowski, 2008), or PredPol (Huet, 2015). Out of the many different aspects of implementing predictive policing algorithms those are selected and discussed in the following that help advancing the concept of PSC regarding the transferability of the concept.

2.3.1. “Police districting problem”

One recurrent topic that is crucial for a successful implementation of predictive policing is the design of patrol sectors given the “police districting problem” (Camacho-Collados et al., 2015). Given that the technology was derived from earthquake prediction systems, locality is key to organize units where to predict possible future events. The performance attributes to develop a meaningful predictive system consisting of workload and response time (Camacho-Collados et al., 2015). The same topic is addressed by Egge (2011) in debating optimized “patrol zones” here under the impression to optimize the use of resources. Egge discussed the design of patrol zones “that could be controlled by a single patrol car” (Egge, 2011: 6). Station infrastructure in this regard is discussed by Skogan et al. (2003). Anticipating the transfer to sustainability control locality is a factor that needs to be addressed. As in policing the legal framework is of major importance, why it is suggested to monitor on a national level embedded in a transnational framework.

2.3.2. Stakeholder involvement

The second major topic derived from reviewed case studies in the involvement of non-police members (Huet, 2015). This is particularly important with regard to PSC, as sustainability affects all members of society and makes stakeholder involvement a necessity. Chainey discusses “police and local partner response opportunities” (Chainey, 2012) and Skogan et al. introduce the “community/business partnership” as one of the three “functional areas” of the CLEAR concept, which is meant to “strengthen problem solving capacities” (Skogan et al., 2003). This is achieved particularly in order to “facilitate easy and convenient information sharing” and secondly, which is the most interesting point for PSC, “intelligence gathering from the community” (Skogan et al., 2003).

For PSC stakeholder involvement also is necessary particularly as the governing body of PSC would not have the authority police forces usually have. Therefore, particularly the involvement of the major stakeholder group of employees (Freeman et al., 2010) in cooperation with non-governmental organizations like the Fair Labor Association (FLA) would make sense.

2.3.3. Cases for expanding the scope of predictive policing: missing data and predictive legislation

Two case studies were dealing with topics that are important for conceptualizing PSC as they cross the boundaries of a functional predictive policing system. As discussed above information gathering also from social media such as Twitter (Wang et al., 2012; Gerber, 2014) is – particularly in the recent years – an increasing factor in strengthening predictive policing. Whereas the integration into the big data pool is based on information that already exists, Stomakhin et al. (2011) make the case for gathering also missing data in order to strengthen the algorithm to predict possible future criminal activities. They take Los Angeles gang networks as an example to discuss the “reconstruction of missing data in social networks based on temporal patterns of interactions” (Stomakhin et al., 2011). The temporal patterns analyzed occur in a “series of interaction events between agents in a social network”, on which the authors base a “reconstruction model that allows one to predict the unknown participants in a portion of those events” (Stomakhin et al., 2011).

A second case study makes the bold step from predictive policing to predictive legislation. The general assumption of Andrade (2012) here is that the “law traditionally reacts after events and is resistant to change and transformation” (Andrade, 2012: 336). Future-oriented technology analysis here is perceived as advancing three specific legal fields: legal research, legislative drafting and law enforcement. The second one “legislative drafting” following Andrade, “presents a new methodological approach to law [...] as a common umbrella term that encompasses foresight, forecasting and technology assessment methods and tools – to the legal sphere” (Andrade, 2012: 351). This “legislative drafting” based on predictive algorithms, “political monitoring” (Lancaster, 2015) and “use of surveillance technologies in planning enforcement” (Harris, 2015) would offer completely new ways for PSC e.g. in the formulation of realistic Sustainable Development Goals: Once a legal basis – be it national or transnational or even global – should be reached, the performance indicators of the SDGs could be used as benchmark for PSC. In that vision public and private data could be used to create a legal enforcement system as it has been described as “digital sustainability panopticon” (Seele, 2016a) including a predictive algorithm pointing at the likelihood of future disasters. However, unlike predictive policing the legal basis is much more complicated, as the SDGs are not a mandatory legal framework but more of an instrument of self-regulation and guidance.

2.4. Evaluation or assessment of techniques or programs

The concept of predictive policing – as promising as it is and as successful as the above mentioned case studies indicate – nevertheless is loaded with expectations and hope, which could be described best with the critical concept of “solutionism” regarding big data as put forward by Morozov (2014). More specifically and in the context of predictive policing Perry et al. (2013) have identified “some myth” about the assumed efficiency of future-oriented algorithms. Hence the evaluation of the success of predictive policing is crucial to think about the transfer of the concept and technology towards PSC.

2.4.1. Measurement and the “ontological distortion effect” of predictions

Before turning to individual studies assessing the efficiency of predictive policing programs one more fundamental and philosophical problem needs to be addressed. Measuring predictions in a precise and accurate way is not possible. This lies in the very nature of the future being unknown – even if predictions may be probable. Ontologically, if predictions do have an effect, a precise measure of the effect is not possible as alternative worlds, in which a meaningful comparison could take place, do not exist. Scientific experiments in standardized lab situations make predictions lead to a situation that does now allow for accurate comparison. Nevertheless, aggregated data does allow to make proximity judgments on the effect of predictive policing in a longitudinal perspective.

If this limitation of what I suggest to call the ontological distortion effect of predictive prevention and control is taken into account research allows at least to compare patterns and arrive at significant results based on correlation, but not on causation (Zwitter, 2014).

Santos (2014) addresses the fundamental question if crime analytics is “cure or diagnosis”. He uses an interesting comparison to ask the question whether the analysis already has an effect on the state of the entity or object analyzed: “Just like the use of MRI results does not cure an illness, crime analysis is the process of using examining data and making conclusions; it is not a crime reduction strategy (cure) by itself” (Santos, 2014: 147). Nevertheless, the underlying assumption of predictive policing is that with the analytical power to predict certain probabilities of future crimes, those events would not necessarily happen, if monitored and patrolled accordingly. Based on this precondition Santos analyzed the use of Compstat and finds that “there is a clear pattern that crime analysis plays a significant role in police approaches that are effective” (Santos, 2014: 147). A second study evaluates different approaches of predictive policing in the context of spatial patterns. Among the approaches compared were point mapping, thematic mapping of geographic areas, spatial ellipses, grid thematic mapping and kernel density estimation (KDE). Chainey et al. find that “KDE was the technique that consistently outperformed the others, while street crime hotspot maps were consistently better at predicting where future street crime would occur when compared to results for the hotspot maps of different crime types” (Chainey et al., 2008: 4). Kuo et al. (2013) evaluate a program for traffic law enforcement. Their finding regarding the effectivity of predictive policing is highly interesting, as their numbers show, that predictive traffic law enforcement at the same time “reduce both crime and crashes” (Kuo et al., 2013: 138). The efficiency here first of all is to be seen in the reduction of “police dispatch time by 13%” (Kuo et al., 2013: 138).

2.4.2. Experimental evaluation

Given the ontological distortion effect of predictive programs Braga (2001) makes use of a different approach testing the effects of predictive policing. Instead of measuring and comparing past empirical data with data under the influence of predictive policing, Braga conducts experiments to test for the effect of predictive measures and the way they channel behavior. Braga identified “five randomized experiments and four nonequivalent control group quasi-experiments” (2001: 104). “The findings of these evaluations suggest that focused police actions can prevent crime and disorder in crime hot spots. These studies also suggest that focused police actions at specific locations do not necessarily result in crime displacement. Unintended crime prevention benefits were also associated with the hot spots policing programs” (Braga, 2001: 104). In sum, predictive policing has a positive effect also in experimental settings and additionally – which could not be

shown by other studies – also unintended crime prevention benefits could be achieved.

Finally and on a more fundamental and general level Byrne and Marx (2011) provide a review on implementation and impact of predictive policing and present in their conclusion three key issues, out of which particularly number 2 and 3 are of utmost importance for PSC: “(1) militarization of crime prevention and policing, (2) coercive vs. non-coercive technology, (3) public vs. private sector control over crime prevention and policing” (Byrne and Marx, 2011: 17).

Point 2 and 3 directly bring us to the discussion of legal and ethical issues around predictive policing. This in regard to the transfer to sustainability will also impose major challenges to PSC as it is not yet clear, who – in practice – would be the right governing authority to establish a PSC.

2.5. Implications and legal/ethical considerations

Big data and here particularly the use of algorithms on large and unstructured datasets represent a disruptive technological change challenging existing structures and infrastructures. Among the greatest challenges is the question, if predictive policing by being predictive is in breach with existing laws and norms. Hence, legal and ethical issues emerge from the preventive character of influencing the future based on correlations between large sets of data.

2.5.1. Legal aspects

As predictive policing is most advanced in the U.S., so is the discussion on a legal level. Several authors address consequences and challenges regarding the Fourth Amendment of the U.S. constitution. Ferguson (2011) states that predictive policing techniques “have constitutional consequences that are only now being considered” (Ferguson, 2011: 179). What is challenged e.g. is the principle of “reasonable suspicion” being one of the pillars of police work. The consequences unfold on two levels. First the “liberty interest of individuals living in those areas [high crime areas], as well as a practical effect on courts analyzing the reasonableness of a Fourth Amendment stop” (Ferguson, 2011: 179). The author’s conclusion from these two points is that the term “high crime area” should be rejected and substituted by a more neutral term. In a subsequent paper Ferguson further elaborates on the Fourth Amendment and the concepts used in that legislation such as “probable cause,” “reasonable suspicion,” informant tips, “drug courier profiles,” “high crime areas”, that have “a significant effect on reasonable suspicion analysis, a reality that necessitates a careful understanding of the technology” (Ferguson, 2012). And in a third paper on the Fourth Amendment Ferguson continues elaborating on the “reasonable suspicion” which he finds to be based on “small data” understood as “discrete facts involving limited information and little knowledge about the suspect” (Ferguson, 2015: 329). Big data instead challenges the traditional paradigm of the Fourth Amendment in so far as “most unknown suspects can be ‘known’ by combining digital surveillance techniques, predictive analytics “combined with biometric or facial recognition” (Ferguson, 2015: 330). Joh (2014) instead finds three uses of big data of relevance regarding the Fourth Amendment: Predictive policing, mass surveillance systems and “potential use of DNA databank samples” (Joh, 2014: 35). Here we can see a fundamental difference between predictive policing and PSC, as the latter is more concerned with legal persons instead of natural persons aka human beings producing a higher complexity by their biological organism compared to legal entities such as corporations.

2.5.2. Ethical aspects

Next to the legal issues of the disruptive changes brought about

by predictive policing and the effect predictions have on the future, also ethical issues are touched upon by predictive policing. Both dimensions – legal and ethical – are of importance in order to understand the potential of predictive policing regarding PSC. Ferguson for example lists a number of concerns that are raised by predictive policing such as: “transparency, accuracy, fairness, equality, and ultimately the legitimacy of GIS crime mapping techniques” (Ferguson 2011: 179). As in legitimacy theory (Suchman, 1995; Seele and Gatti, 2015) next to the legal ‘license to operate’ there is the notion of legitimacy that also can be challenged. Here predictive policing offers new horizons that challenge the overall legitimacy of the concept. Casady (2011) is even more blunt in discussing the ethical issues of predictive policing and sees overall ‘police legitimacy’ at stake. Due to predictive policing “ethical problems for police, especially regarding the perception of racial profiling and a perceived lack of procedural justice” is created (Casady, 2011). Brakel and Hert (2011) discuss a series of “unintended consequences” of predictive policing, which are exemplified by a series of judgments of the European Court of Human Rights, addressing what Perry et al. refer to as “civil liberties and privacy rights” (2013).

2.6. Critique and pitfalls of predictive policing

Predictive policing, as promising (Perry et al., 2013) as it is regarding crime prevention, and as technologically disruptive as it is regarding legal and ethical issues, poses positive evaluations among others. This last part of the review goes beyond legal and ethical concerns raised above and refers to a publication of Ridgeway (2013) in the U.S. National Institute of Justice (NIJ) presenting seven possible perils of prediction in the context of policing. Ridgeway – acting director of the NIJ – presented the research to U.S. law enforcement agencies developing predictive policing programs. Hence, the worries are less from a fatalistic concern but are addressed at the very heart of the concept of predictive policing. The seven pitfalls are meant to adjust expectations and also address the issue of “biased predictions” and the “plagued persistent shortcomings” (Ridgeway, 2013: 34).

For the development of PSC the shortcomings and inappropriate expectations represent a valuable help in designing PSC in a way that leads to meaningful improvements regarding the achievement of sustainability.

The first pitfall mentioned is labeled: “*Trusting Expert Prediction Too Much*”. Ridgeway by an example from medical science and Supreme Court’s judgments shows that trusting (human) experts and their predictions is outperformed by predictions of algorithms. Hence pitfall 1 indeed is less a pitfall of predictive policing, but a strong point for having even more predictive policing. Transferring this finding to sustainability we may say that the future prospects of e.g. the climate conferences also were trusted too much.

Pitfall two addresses blind trust in basic statistics of model assumption and hypothesis testing (as also known in most of quantitative academic research): “*Clinging to What You Learned in Statistics 101 If your knowledge of prediction.*” Here Ridgeway speaks about the new technological basis of algorithm based predictions that outperform the classical way of testing for statistical significances in static models. The key to make the paradigm shift clear is the accuracy of predictive algorithms along with its complexity of incorporating, e.g. also data of working hours needed from a police officer. Another criterion highly important for the PSC concept is the “interpretability” of the data, which means that a person can “understand why the prediction model makes the predictions it does” (Ridgeway, 2013: 37). Pitfall three: “*Assuming One Method Works Best for All Problems*”. As for PSC the complexity of indicators and data requires a multitude of different methods to be combined.

Big Data also means ‘big methods’ as we can derive from this pitfall. Pitfall four addresses the scope: “*Trying to Interpret Too Much*”, which again addresses the overrated expectations of experts in comparison to the accuracy of the algorithm. Pitfall five: “*Forsaking Model Simplicity for Predictive Strength — or Vice Versa*”. Here Ridgeway argues that interpreting results “is more important than achieving greater predictive capability” (Ridgeway, 2013: 38). Pitfall six again is very important for the concept of PSC: “*Expecting Perfect Predictions*”. For predictive policing we find the same goal that would be adopted for the PSC: “Predictions will not be perfect, but the ultimate goal is to improve overall efficiency” (Ridgeway, 2013: 38). Finally, pitfall seven addresses unintended consequences, as they have already been addressed in the ethical issues above: “*Failing to Consider the Unintended Consequences of Predictions*”. Here Ridgeway addresses similar concerns as mentioned above, particularly given the nature of statements regarding the future that did not happen so far. This ‘as if’ temporality might cause unintended consequences challenging the success of the predictive policing concept.

Regarding PSC, the seven pitfalls also indicate the limits and challenges of the promises predictive analytics could bring to sustainability. In the following those elements of predictive policing that can be utilized for developing PSC are identified and conceptualized.

3. Transfer: what corporate sustainability control and management can learn from predictive policing

To organize the conceptual transfer from predictive policing as presented above in the literature review to the idea of PSC, the new concept is developed on different levels: First a conceptualization and definition is derived from the criteria mentioned in the review. Secondly the operationalization of the concept is discussed building on existing data management technology in corporate data management and reporting (XBRL) as well as on the future ICT concept of a “planetary nervous system” (Helbing, 2012). In this regard PSC is part of a young research field on combining big data and sustainability (Song et al., 2016; Du et al., 2016; Seele, 2016a,b; Zhao et al., 2016).

3.1. Conceptualization of PSC based on the review

The transfer from predictive policing to sustainability however does not go without conceptual challenges. Policing is an activity mandated by public authorities for law enforcement. Sustainability instead is a transformative concept having a normative core as it aims for a transition (McCormick et al., 2013) in the future. In the following the core parameters of predictive policing as identified in the review are summarized and transferred to the field of sustainability. The first column of the transfer-table indicates the references from the review on predictive policing as presented above. The second column summarizes features of predictive policing. The third column finally proposes a transfer of predictive policing features and how they could be used to develop PSC. It needs to be said however, that the third column presenting the features of PSC are developed by structural analogy and not by empirical evidence. The features instead are meant to stimulate debate and empirical validation of the PSC concept.

Based on the review the core elements of PSC are outlined. The criteria as derived from policing help in developing the application context of the concept.

In the following PSC is conceptualized also on the level of definitions in order to arrive at a concept that could be debated, advanced and modified. Reviewing the literature on predictive policing, the most concise definition of predictive policing stems

from Perry et al. (2013) which I will adopt for developing a definition of PSC. However, as PSC is not an authority like the police, the definition builds on the idea of deliberation between companies, stakeholders, NGOs and regulators. Therefore, it reads:

Predictive Sustainability Control is the use of analytical techniques to identify subjects for mutual deliberation, supervision and intervention with the goal of preventing future harm related to environmental, social and governance issues, solving past scandals, and identifying potential actors/corporations of unsustainable activities and their stakeholders in the near future.

An extension of the definition can be adopted from the definition of the predictive policing program CLEAR (Skogan et al., 2013) defining three functional areas. Hence the definition of PSC can be explained further:

Functional Differentiation: PSC applications in a corporate context impacts three major functional areas: i. sustainability management, ii. stakeholder partnership and iii. regulatory integration. In the area of corporate sustainability management, PSC should promote effective resource allocation, human resources management and accountability, risk management and early warning, tactical and strategic planning, and fiscal accountability. In achieving regulatory integration, PSC will facilitate the development and implementation of unified strategies to reduce future harm, eliminate sustainability strategy “bottlenecks,” increase accountability among regulating agencies, and provide a comprehensive profile of offender activity. In the area of stakeholder partnership, PSC should strengthen problem solving capacities, conduct community needs assessments, and facilitate easy and convenient information sharing and intelligence gathering from the community.

3.2. Operationalization: XBRL in a “planetary nervous system”

If – on a conceptual level – the development of a PSC system would be an option to reach and enact the transformative approach of sustainability (Mader, 2013; McCormick et al., 2013 or Higgins, 2013), the next step is to operationalize the concept. This can be reached on different levels as the following figure indicates, which is explained in detail below:

The most general level to operationalize PSC is the given of the so-called “internet of things” (Greengard, 2015), where things and machines are connected to each other and where things are also connected to human beings. On the level of the internet of things big data is gathered. Visionary scholars such as Helbing developed based on the technological possibilities the concept of “Future ICT” (Helbing, 2012) or subsequently, a big data driven “planetary nervous system” (Helbing, 2014). This is in line with other scholars such as Gijzen, who made a famous claim in the journal Nature for utilizing big data for a sustainable future (Gijzen, 2013). The most advanced macroscopic approach however can be seen in the planetary nervous system “as citizen web” (Helbing, 2014), which aims to create an “open, public, intelligent software layer on top of the “Internet of Things” as the basic information infrastructure for the emerging digital societies of the 21st century” (Helbing, 2014). The architecture of the “Planetary Nervous System” would be open source (similar to Wikipedia or Linux) and first attempts are already under construction (see @NervousNet or the App “Global Participatory Platform”). The basic points, that matches well with the PSC concept is the increased acceleration of innovation in order to address and possibly solve the challenges and obstacles that put a thread to the contemporary world.

As a next step the internet of things and planetary nervous system when used as a superstructure for PSC, would need a link to the business world and the environmental, social and financial data produced at corporations and supervised by regulators. As algorithms in big data also are capable of analyzing ‘unstructured data’ already existing performance indicators and measurements would however help in advancing and speeding up the development due to the increased usability of standardized data. Here XBRL comes into play. XBRL stands for “eXtensible Business Reporting Language” and is a digital business reporting standard initially created for financial reporting and since 2009 made mandatory as reporting format by the U.S. Securities and Exchange Commission (SEC). It allows for real-time transparency and reduced data-migration errors and transaction costs (Burnett et al., 2006).

Most important for the operationalization of PSC are first attempts to adopt XBRL to Corporate Social Responsibility (CSR) and corporate sustainability data and metrics (Harris and Morsfield, 2012) and collaborations with GRI 4.0 have been established (Knebel and Seele, 2015). XBRL has led to the idea of a real-time transparent reporting format for ESG data called “Digitally Unified Reporting” (Seele, 2016b) paving the way for regulators like the SEC or others to also monitor and regulate sustainability performance and to arrive a more credible reporting practice (Lock and Seele, 2016). In addition, by reporting in XBRL the latest trend in CSR reporting – incorporating earth as finite – could be taken into account and controlled for rigorously (Bjørn et al., 2015).

Another cornerstone of PSC could be seen in footprint analysis tools (Čuček et al., 2012), providing data bases and usage statistics that could be fed into the PSC concept.

In a final step, the predictive analytics as already approved in predictive policing could be applied to the structured and unstructured data to arrive at a functional PSC. Here, building on the definition from the previous chapter, the three functional areas i. Sustainability management, ii. Stakeholder partnership and iii. Regulatory integration help making the PSC become reality (for the entire operationalization process see Fig. 1 above) (See Table 1).

3.3. Governance: need for a world-sustainability-police or business incentives to enforce PSC?

The main challenge for developing PSC is not the technology and also not the process design and operationalization for the architecture of PSC, but the governance structure to bring to bear and apply the concept of PSC. Given the little success of the international community in achieving previously defined sustainable development goals and reduction of emissions, it is considered unlikely that a global authority governing PSC would be agreed upon. The PSC could alternatively be mandated by or commissioned of the United Nations, being interested in the topic for many years as e.g. the Global Compact indicates.

However, as PSC is based on the functionalities of predictive policing, the governance of PSC would lead to an (uni- or multi-lateral) authority, that if rigorously controlling and enforcing sustainable development, would possibly lead to the creation of something like a sustainability police. This in return raises questions on the foundations and the sincerity of sustainable development. If it is meant not only to greenwash unsustainable practices enforcement would be an issue.

As the author is skeptical about the success and appropriateness of a quasi-totalitarian authority governing the process or reaching sustainability (as unsuccessful other ways might indeed be), other ways of obtaining the sincerity of PSC without creating a quasi-totalitarian authority should be discussed (Seele, 2016a). In the following three different ideas to promote PSC in a voluntary way are discussed: *Fiscal Incentives and Trade Barriers, Access to Capital,*

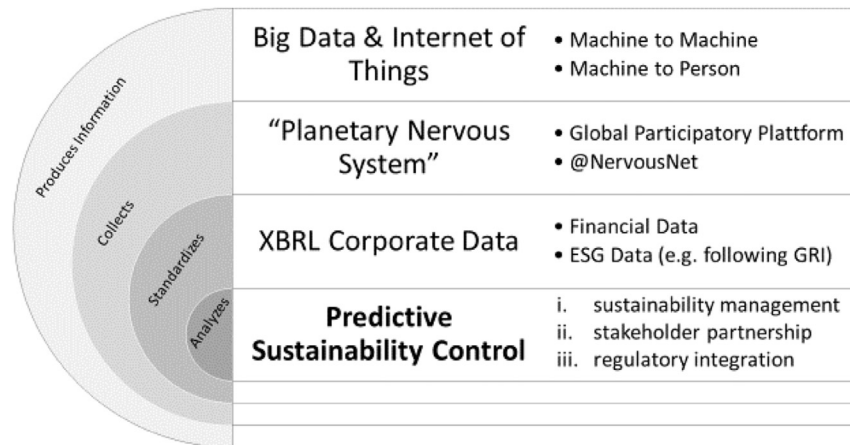


Fig. 1. Operationalization of Predictive Sustainability Control: From the internet of things down to standardized taxonomic data points to predictive analytics.

Insurance bonus-malus schemes and consumer responsibility PSC derived labels.

3.3.1. Fiscal incentives

Public authorities making use of PSC data and predictions could develop a scheme including fiscal incentives for companies participating voluntarily and contributing their unstructured and structured XBRL data to the predictive analytics. This would give an incentive to companies on the basis of a trade-off whether or not to participate. First mover incentives occur as well as critical mass phenomena, meaning if a certain number of companies join, the other would follow. This could be similar to the CSR movement, which by now is almost impossible not to address by a corporation. Next to fiscal incentives, also trade barriers could be discussed. Strong economies on the supply or on the demand side could ask for a PSC participation or certification in order to get market access (similar to the forest stewardship council for sustainable wood products).

3.3.2. Access to capital

Similar to pension funds or sovereign wealth funds applying ethical investment guidelines or UN's principles for responsible investment, private or public banks or funds could make participation at the PSC system a criteria for access to capital. In addition, here the gaining momentum principle would be the driving force as single investors or funds although subscribing to the PCS idea, would not make a difference, if the majority of businesses keep maintaining the current modus operandi.

3.3.3. Insurance bonus-malus schemes

Unsustainable corporate activity could happen as a scandal with a peak in media attention or – which is the majority of cases – in systematic business as usual standard procedures producing emissions and social unease way beyond what would be tolerable in sum to speak of a sustainable world. As national regulators – as long as relocation to circumvent environmental or social standards is the case – are limited in scope, the consequences of unsustainable production also affects the future of businesses. Here insurance companies covering both scandals and disasters as well as long-term effects such as climate change, could come up with bonus-malus schemes.

3.3.4. Consumer responsibility by PSC labels

Alternatively, PSC participation could lead to a label or certificate. Than it would also be in the decision of consumers to opt for or

against the rigorous application and prevention of unsustainable activities. Organic labels have shown that consumers have a higher willingness to pay, and at the same time labels have the power to transform business practices. This so far however was voluntary or has become legislation, but did not have an overall effect on reaching sustainability. PSC labels would – unlike existing labels – not be an indicator or auditing based label but a label indicating membership in the “global participatory platform” mentioned above, that is part of the operationalization of PSC also on a technological level. Hence, it would be a label more for indicating membership in a new data driven world also pushing sustainability goals, instead of ticking boxes of indicators highly debatable if they really promote sustainability.

4. Future research: theory implications and empirical testing

The proposed concept of PSC bears implication also for theory development. These are touched upon here only briefly and invite for further research and empirical testing.

As big data and the internet of things need to be seen as technological disruption brining about a major change, this disruption also affects our understanding of concepts and theories regarding sustainability. As some argued big data means “the end of theory” (Anderson, 2008) given the analytical power of big data and pattern recognition, it is argued here, that big data and predictive analytics as applied in PSC open a new chapter in theorizing on CSR and sustainability: Existing theories on CSR can be classified along different lines, as they vary immensely regarding their scope and implications. In sum there are two opposing approaches that can be characterized as ‘instrumental’ (strategic CSR or Creating Shared Value) and ‘deliberative’ (Habermasian Political CSR) (for a systematic review see Seele and Lock, 2015): On the one hand there is strategic or instrumental CSR and corporate sustainability, where responsibility and sustainability is used to make profit by increased willingness to pay or brand equity. On the other hand, we find normative or transformative approaches putting social responsibility and sustainability in the first place. So far no theoretical approaches exist addressing and incorporating the paradigmatic changes of big data and digitalization. Here PSC would be a suitable case to develop theory further and advance our understanding of corporate sustainability in the digital age.

Regarding sustainability theory (without a primary focus on business) discussions are centered around the sustainability discourse (Christen and Schmidt, 2012) or on the ‘grand theory’ of capabilities. Burger and Christen (2011) developed a capability

Table 1
Conceptualizing Predictive Sustainability Control derived from the review of predictive policing.

Reference on predictive policing	Features of predictive policing	Transfer of features for Predictive Sustainability Control
Overall efficiency: (Kennedy et al., 2011, Greengard, 2012; Ridgeway, 2013: 38)	"Predictions will not be perfect, but the ultimate goal is to improve overall efficiency" (Ridgeway, 2013: 38)	PSC is to improve overall efficiency in controlling and achieving global sustainable development goals.
Focusing on the future (Beck and McCue, 2009: 18)	"moves law enforcement from focusing on what happened to focusing on what will happen and how to effectively deploy resources in front of crime, thereby changing outcomes" (Beck and McCue, 2009: 18).	As in sustainability reporting, which is faced backward, PSC is focused on real-time-transparency and predictive analytics to identify future harms and reach sustainability goals.
'predictive mapping' (Chainey, 2012)	'predictive mapping' has been "extended to other types of crimes such as pedal cycle theft, vehicle crime and street robbery" (Chainey, 2012)	Like predictive policing PSC is clustered base on spatial data, here on national and district level. Subsequently it could be extended to other types of unsustainable activities.
"decision support system" (Camacho-Collados and Liberatore, 2015).	Predictive policing on the contrary with the help of big data and predictive algorithms, allows probabilistic pattern recognition to predict future risks	PSC also can be seen as 'decision support system' allowing better judgments to prevent future risks.
Crime Patterns (Greengard, 2012: 19–20).	utilizes data concerning the time, distribution, and geography of past criminal activity into a database to determine crime patterns, which allows police to adjust patrols and resources to improve crime deterrence and better apprehend subjects"	PSC also makes us of existing empirical data from the past to better allocate supervision and control and also to develop patterns of unsustainable corporate activity
Spatial dimension "places, not people" (Curry, 2015),	"places, not people" (Curry, 2015). Spatial distributions however offer predictive models to make person-independent forecasts, that make police work more effective, also on the level of backup covering (Curtin et al., 2010): "risk terrain modeling" compared to retrospective mapping approaches over six months produced significantly more reliable predictions (Caplan et al., 2011).	Person-independent or in the case of PSC company-independent forecasts help making promotion of sustainability more effective.
Temporal dimension Cohen et al. (2007)	"risk terrain modeling" compared to retrospective mapping approaches over six months produced significantly more reliable predictions (Caplan et al., 2011). Cohen et al. (2007) developed a micro level scale of a one month ahead forecast. I	Also in PSC risk as a factor to evaluate certain terrains is chosen in order to arrive at more reliable predictions
Spatio-temporal Models	Twitter messages are often "tagged with precise spatial and temporal coordinates" and thus allow to automatically identify discussion topics across a major city" by linguistic analysis and "statistic topic modeling (Gerber, 2014: 115). The model from Wang et al. (2012) is called the "spatio-temporal generalized additive model (STGAM)" (Wang et al., 2012	Predictive temporal horizon for PSC is suggested to be similar to poling, however sustainability specific time horizons are required to be developed. Twitter-based spatio-temporal modeling incorporating tagged geographic and demographic data, would be highly relevant to identify particularly socially unsustainable behaviors, see the Foxconn example in the introduction.
Process Design (Kuo et al., 2013: 138).	1) geocoding the data 2) defining the hotspots 3) organizing the best patrol routes 4) estimating the effectiveness	For PSC the procedures suggested by Kuo et al. in general make sense. Starting with geocoding is useful as certain regions are more sensitive to socially or environmentally unsustainable activities than others. This could lead to some hot spots such as deep-sea drilling, carbon emissions, collective suicides, or large-scale human rights violations, just to mention a few candidates for hot-spots. As PSC is not governed by a police authority, some alterations would have to be applied. Here the attention of NGOs or regulators could be organized along the results from 1 to 2. As predictions change the future, measuring of effects is complicated. Hence the overall reaching of sustainability goals based on the established clusters would be the measure for effectiveness.
PredPol adopted the earthquake prediction software by following two types of variables. [...] Each factor can be boiled down to the usual rate it triggers other crimes" (Huet, 2015: 46).	Fixed Parameters (gang shooting triggering retaliation) Variable Parameters Upcoming paydays (here budget constraints or profit targets)	Fixed parameters for PSC could be the known and systematic violation of human rights and environmental standards. Other parameters could be the risk-disposition of certain industries as e.g. in the chemical industry. Variable parameters instead could be the announcement of a new and higher profit-target of a company triggering activities on the cost of employees or the environment. Other variable parameters could be e.g. economic crisis or scandals shaking an entire industry.
Profiling of Individuals (Duhigg, 2012)	tracing habits and changes in habits	As some companies seem to be more unsustainable than others and produce more scandals than others, the profiling of suspect companies could be an option. Monitoring their habits and also their

Table 1 (continued)

Reference on predictive policing	Features of predictive policing	Transfer of features for Predictive Sustainability Control
context data	not only data of the individual person from past and present is used, but also context data regarding neighborhood, social network, or previous communications with others.	publications on sustainability could help identifying their risk profile. Here e. g. changing legislation in countries with high relevance could be monitored to predict future risks. If e.g. a new emission guideline is issued, companies might react in different ways to the new norm.
Multi-Methods (Gorr and Harries, 2003: 551).	geo-data later on was extended by “long- and short-term horizons, univariate and multivariate methods, and fixed boundary versus ad hoc spatial cluster areal units for the space and time series data” (Gorr and Harries, 2003: 551). Among the approaches compared were point mapping, thematic mapping of geographic areas, spatial ellipses, grid thematic mapping and kernel density estimation (KDE).	PSC also would benefit from long and short term time horizons, particularly regarding peaks like scandals and long-term goals as emission reduction or waste management.
“spatio-temporal generalized additive model” (Wang et al., 2012)	They developed a spatio-temporal modeling making use of geographic, demographic and Twitter-derived information. The model they develop is called the “spatio-temporal generalized additive model (STGAM)” (Wang et al., 2012)	Thematic mapping seems to be particularly suitable as e.g. GRI Indicators could be monitored differently (see next chapter on XBRL)
Cross communication (Chainey, 2012)	“police and local partner response opportunities”	STGAM also is a suitable model for PSC, particularly as it incorporated spatio-temporal data from Twitter.
Cross-Partnership (Skogan et al., 2013)	Skogan et al. (2013) introduce the “community/business partnership”	As sustainability heavily depends on stakeholder involvement, a specific communication tool for exchange between corporations, regulators, employees and civil society is highly necessary. Involvement of the major stakeholder group of employees (Freeman et al., 2010) in cooperation with non-governmental organizations like the Fair Labor Association (FLA) would make sense.
Public Policy Level	This “legislative drafting”. (Andrade, 2012: 336). “political monitoring” (Lancaster, 2015) “use of surveillance technologies in planning enforcement” (Harris, 2015)	Given the attempts to reach sustainability via goals of the international community, predictive legislative drafting seems to be a helpful step in achieving de facto sustainability goals. As corporations mainly react to legislation, political monitoring as in predictive policing is an appreciated tool in finding solutions to actually arrive at a (more) sustainable world. A mere trust based sustainability does not seem to lead to sustainability, surveillance technology also in PSC is considered a helpful technology to prevent future harm.
Pitfall (Ridgeway, 2013: 38)	“Failing to Consider the Unintended Consequences of Predictions”.	As PSC would be a pilot, also here unintended consequences could occur. Hence monitoring about success and possible loopholes or damages is needed right from the beginning
Promises and Myths	policing Perry et al. (2013) have identified “some myth” about the assumed efficiency of future-oriented algorithms	The skepticism towards predictive policing and its actual success is justified as we do not know the future without predictions. Hence also PSC should be looked at as a promise, monitored closely.
Testing effects of predictive analytics	Braga identified “five randomized experiments and four nonequivalent control group quasi-experiments” (2001: 104)	Given the skepticism mentioned above it would be highly recommended to also conduct tests to analyses the psychological effects of being under surveillance and subject to predictive analytics.
Legal and Ethical	“have constitutional consequences that are only now being considered” (Ferguson, 2011: 179). Privacy: predictive analytics “combined with biometric or facial recognition” (Ferguson, 2015: 330). “transparency, accuracy, fairness, equality, and ultimately the legitimacy of GIS crime mapping techniques” (Ferguson 2011: 179) Profiling and discrimination: ethical problems for police, especially regarding the perception of racial profiling and a perceived lack of procedural justice is created (Casady, 2011). Assumption of innocence	One possible unintended consequence could be a transformation of the deliberative resources in a society affecting the power of critical thinking. Most companies being private legal entities would also face challenges regarding their privacy. The overall legitimacy for PSC needs to be researched and discussed just as for predictive policing. Unjustified accusations or discriminations due to PCS because of past unsustainable behavior would be just as inappropriate as for predictive policing. As in predictive policing the assumption of innocence is to be upheld, even if predictive analytics indicate to the high probability of a future activity. Nevertheless, activities of ‘checks and balances’ as well as NGO involvement would make sense to prevent future harm without making unjustified accusations.

approach of sustainability in this journal. This approach originates from A. Sen builds on the basic assumption of what individuals are able and capable to do. PSC could be theoretically subsumed under the capability approach and the six criteria of Burger and Christen would be a suitable framework to develop a capability approach driven theory of digital sustainability. More recent approaches (Schultz et al., 2013) of the capability approach for sustainability also address employee feedback. This would match well with the second functional area of PSC being stakeholder partnership with employees being key stakeholders.

Overall the amount of overarching theories covering the changes of the digital age still are missing. What seems to be promising is the discourse on real-time transparency, big data correlations without focusing on the individual and last but not least theories of privacy in the digital age. PSC as presented here would be an application context that might also serve for theory development. Further elaborations on theoretical implications would go beyond the scope of this article, but require future research for theoretical advancement and empirical testing of the behavioral effects of PSC.

5. Limitations and the discursive value of PSC to stimulate debate

PSC as developed here is not a blueprint or architectural design for developing a generic algorithm to predict environmental and societal disasters. Instead – at least at this first stage – it is a theoretical concept derived from a literature on predictive policing and transferred to the arena of cleaner production and sustainable development. The review has focused on the advancements that predictive policing brought to law enforcement and – by analogy – transferred the parameters on a technical and procession level to the field of sustainability. Hence PSC is a shift of the application context of predictive policing making use of the already established predictive analytics based on big data in the field of forecasting future unsustainability and reducing overall harm regarding ESG criteria. It can be assumed that on a technical level PSC would work as efficient as predictive policing, once the set up and data collection and preparation is in place. The largest limitation, also discussed above in the chapter on governance, however seems to be the legal framing and governance structure of PSC. In sum the main limitations can be found:

- Who is in charge of PSC? A mandate for an international organization like UN? Legal frameworks of single nation states? Also a generic software program could be developed for use in corporations or along specified supply chains.
- Predictive policing enforces the law within an existing legal framework. Sustainability is soft law – at best. Based on what normative criteria will the measurement and prediction of future events be guided?
- Predictive policing is consistent within existing legislations. Sustainability however is a global phenomenon and actors are transnational legal entities such as corporations. Relocation to less regulated environments remains an option hardly covered by PSC.
- Resources. Although many international and national organizations with budgets are dedicated to promote sustainability, it is not easy to imagine a coordinated action to promote PSC on a global level.
- Sustainability itself is a contested concept without a binding definition. PSC would need a unified definition of sustainability to code the algorithms accordingly as sustainability could be reached only on a global scale, given phenomena as floods,

climate change and global warming not stopping at national frontiers.

In the light of the severe limitations to start immediately a program of PSC, the value of the concept needs to be defined more specifically: Given the current unsustainable world we live in co-ordinated action is needed – just as mentioned above by Obama when saying that even the U.S. and China cannot do this alone. PSC is a concept that provides a technologically driven way out by building on big data and predictive analytics in a digital 'planetary nervous system'.

Hence the value of the PSC concept – at the moment – is to be seen more in starting a debate than programming a code. The debate as triggered by PSC therefore challenges the 'business as usual' sustainable development with aspirational talk and conferences and voluntary development goals, as on a technological level we do have the means to arrive at a sustainable world. This can be shown by the availability of the PSC concept. And given the progress in global unsustainability, this debate although it would restrict freedom and have elements of a police state in the name of sustainability, shows a big data enhanced way of overcoming scandals and unsustainable practices.

6. Revisiting the thought experiment of predicting scandals

Given the limitations and obstacles in making the concept of PSC real, it is important to outline, what the specific contribution of the concept would. To create awareness for the need for a big data based predictive system the article started with a thought experiment to prevent major sustainability disasters such as Rana Plaza building collapse in Bangladesh killing 1127 workers or the Deep Horizon oil spill.

The concept as developed here at the current state is not a remedy for the future occurrence of these kind of disasters even if all the obstacles mentioned above were removed. However, in the following it is outlined how a PSC system could be designed to point in the direction of predicting and preventing future disasters. In the case of Rana Plaza the data processing could be conceptualized as follows: Along the data-operationalization concept (Fig. 1) unstructured big data is collected by the internet of things. This data could stem from mobile phones of the employees as well as from machine to machine communication taking place at the company. This unstructured data then is fed into the 'planetary nervous system'. Here sensors and cameras of employees could collect data. Also public data regarding geographical parameters could be used, bringing back the geo-tagging origin of predictive policing also into PSC. This geographical data could be used to identify patterns that e.g. in Bangladesh there were several problems with constructions, earth quakes and corruption. If this data is combined with the industry (here apparel and textile) this could lead to risk assessment predicting future events. The different data points than could be used by a PSC system on behalf of the supply company or the consumer company selling the products. As mentioned above an XBRL database could be used to administer all data points in a standardized was. Hence single companies would not have to develop their own PSC system, but a generic programming would allow for a standardized solution. With the XBRL taxonomy data from environmental and social performance as used in CSR reporting, structured along the lines of GRI key performance indicators than could be used to feed the PSC system at company level. Finally, the generic algorithm of a PSC would involve

- corporate data from sustainability management
- public data from stakeholders like NGOs, consumer advocate groups, regional or public authorities, and

- regulatory integration based on the specific legal framework (as long as there is no global governance for ESG) of the company and if required their clients.

As in predictive policing the system works on clustering past data and building probabilities of future events. Therefore, in the development of PSC a database of past scandals affecting sustainability is crucially important for reliable predictions. Furthermore, the PSC as indicated in the data operationalization (Fig. 1) needs involvement of general stakeholders and secondly of legal authorities, as predictions may challenge the burden of proof and assumption of innocence. In sum however the disadvantages as reputation damage or risk discrimination are to be seen against the potential advantages of preventing major scandals and disasters and even more so establishing a big data bases system of monitoring and surveillance possibly saving thousands of lives and reducing harm to the environment.

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